

SUMMER INTERNSHIP - 2026

AI Agents for Digital Capability Building in Tata Steel

WEEK-4

Make It Intelligent (Data + Context + Scale Thinking)

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From Demo to Real-World Solution

Transitioning From Demo to Production

In Week 4, the primary goal was upgrading the simple prototype chatbot into an enterprise-ready, context-aware digital learning assistant. We integrated real and simulated company documents, implemented safety access controls, and improved query retrieval quality.

💡 *Mentor Push:*

*"Would you trust this AI in a real training environment?"
To meet this standard, we targeted zero-hallucination vector searches, robust RBAC safety blocks, and 100% accurate contextual grounded Ness.*

1. Domain-Specific RAG Database

Indexed leave, onboarding policies, safety codes, bearing manuals, NSDC roles, and Tata reports into separate, dedicated vector collections.

2. Context-Aware Personalization

Integrated SQLite DB containing employee profiles (name, current/target role, readiness score) to tailor learning recommendations instead of generic advice.

3. Robust Guardrails & RBAC Security

Introduced strict user types (Employee, Manager, etc.) to block access to production data, returning professional block warnings instead of unauthorized info.

Multi-Agent System Architecture

Specialist Agent Orchestration

Instead of a single large LLM prompt, we deployed five specialized, dedicated agents coordinated by a central Orchestrator. This ensures separation of concerns, high accuracy, and isolated database contexts.

Orchestrator Routing Flow:

1. User input query received.
2. LLM Intent Classifier classifies category.
3. Role-Based Access Control (RBAC) checked.
4. Routes query to the designated specialist agent.
5. Returns structured answer and verified document sources.

Central Orchestrator

Classifies query intent (KNOWLEDGE, HR, MAINTENANCE, LEARNING, PRODUCTION).

Enforces RBAC access control matching user types.

Knowledge & SOP Agent

RAG: Searches safety manuals, NSDC roles, annual and sustainability reports.

Utilizes query expansion fallback.

HR Support Agent

RAG: Searches leaves, benefits and policies.

HyDE-lite translation for Hinglish/Hindi, query refinement loop.

Maintenance Agent

RAG: Equipment manuals and codes.

Direct role-name detection, tiered fallback searching.

L&D Learning Agent

SQL: Directly queries SQLite database.

Extracts readiness score, missing skills, lists courses, dynamic tasks.

Production Agent

SQL: Aggregates downtime and KPIs from DB.

RBAC restriction: Restricted to Managers and Executives.

Specialist RAG Agents & Design Rationale

Knowledge & SOP Agent

Scope: Company policies, SOPs, safety manual, annual reports, NSDC roles.

Architecture Highlights:

- Targeted Collections: Maps keywords to target specific collections, preventing cross-domain overlap.
- Query Expansion: If the initial query returns no matches, an LLM rewrites the query with synonyms/financial terminology.
- Clean Source Reporting: Only lists PDF sources if the final generated response actually utilized the context.

HR Support Agent

Scope: Leaves, payroll, onboarding, probation, benefits, HR policies.

Architecture Highlights:

- HyDE-lite Query Translation: Translates input query to English for embedding search to match English documents.
- Output Match: Responds in the exact input language (Hindi, English, or Hinglish) using tailored prompts.
- Sufficiency Check: Employs a self-RAG LLM sufficiency check; expands and retries if context is lacking.

Maintenance Agent

Scope: Machine vibration, sensor alerts, LOTO procedures, bear operating range.

Architecture Highlights:

- Tiered Search Flow: Searches specific `maintenance` collection first. If empty, falls back to `role\_knowledge` collection.
- Role Matching: Scans the query for specific plant roles (e.g. Heating Regulator) and filters search to that metadata field.
- Help Desk Escalation: Falls back to Maintenance Supervisor (Ext. 301) for unmentioned safety procedures.

Database-Connected SQL Agents

L&D Learning Agent

Objective: Access personalization and training career roadmaps.

Architecture & Implementation Details:

- Database Integration: Gathers data from SQLite `users`, `skill\_gap` and `recommendations` tables using user_id.
- Personalized Context: Extracts user metrics like Experience Years, Target Role, Readiness Score, and Missing Skills.
- Dynamic Task Customization: Checks query keywords to customize prompt focus. For example, queries about duration prompt a focus on recommended course durations and priority order.
- Privacy Protection: Restricts recommendations to the logged-in user only, blocking cross-employee data leaks.

Production & KPI Agent

Objective: Provide managers/executives with plant efficiency data.

Architecture & Implementation Details:

- Structured SQLite Queries: Queries Unit KPIs (e.g. Unit 3 underperformance) and aggregates downtime reasons (Raw Material Shortage, Power Outage, Equipment Failure).
- RBAC Access Control: Integrates directly with the central Orchestrator. The orchestrator checks if the current user_type is 'Manager' or 'Executive' before routing to this agent.
- Secure Rejection: If an Employee or New Joiner asks about Unit downtime, the query is blocked, returning a friendly refusal page.

Development Challenges & Solutions (Part 1)

Challenge 1: Multilingual Retrieval Performance

Problem:

Ground-level plant technicians and onboarding employees often search in Hinglish (Hindi written in Roman letters, e.g., 'mujhe kitni casual leave milti hai') or Hindi. Searching directly in English policy documents yielded high vector distances, retrieving irrelevant sections or causing RAG failures.

Mitigation Strategy: HyDE-Lite Normalization

1. Added a translation layer using Llama-3.1-8B.
2. Pre-processed Hinglish/Hindi queries into clear English search queries for vector similarity matching.
3. Conducted the final LLM response generation using the original question, ensuring responses match the input language.
4. Output: Reduced vector distance from 0.56 to 0.29 (47% improvement).

Challenge 2: Cross-Domain Chunk Leakage

Problem:

Vector search across a global database returned unrelated documents due to overlapping keywords. For example, a query about 'welding safety' retrieved HR policy snippets instead of safety manuals because of generic corporate vocabulary.

Mitigation Strategy: Semantic Targeting

1. Reorganized database into separate collections: ``knowledge_safety``, ``knowledge_reports``, ``role_knowledge``, ``maintenance``, ``hr_support``.
2. The Orchestrator routes the search to specific collections based on classified intent.
3. Role Detection: Matches names (e.g. Heating Regulator) in the query and filters search results accordingly.

Development Challenges & Solutions (Part 2)

Challenge 3: Low Grounded Ness & Hallucinations

Problem:

The agent evaluation report initially showed an Average Grounded Ness/Faithfulness score of just 20.0%. The LLM commonly cited irrelevant PDFs or hallucinated numbers that were not present in the vector DB chunks.

Mitigation Strategy: Self-RAG & Post-Filters

1. Sufficiency Loop: Built a Self-RAG validation check. An LLM judge evaluates if retrieved chunks are sufficient. If not, the query is refined and resubmitted.
2. Deduplication Filter: Deduplicates text blocks, preserving the highest-ranked embeddings to fit clean contexts into the LLM context window.
3. Source Utilization Validator: A prompt filter checks if the LLM couldn't answer from context. If it answers 'not found', it clears the cited PDF sources.

Performance Evaluation Summary

Using LLM-as-a-judge (Llama-3.1-8B-instant) on standard test suites:

- Intent Routing Accuracy: 100.0%

The orchestrator routes KNOWLEDGE, HR, LEARNING, PRODUCTION queries with 100% accuracy.

- RBAC Safety / Block Accuracy: 100.0%

Employees and New Joiners are successfully blocked from accessing sensitive Unit downtime details.

- Answer Relevance Score: 90.0%

Answers directly target the employee's intent.

- Average Answer Correctness: 60.0%

Factual correctness evaluates to 60.0% matching expected ground truth answers.

Search Distance Improvements (Hinglish vs English)

Embedding Distance Analysis

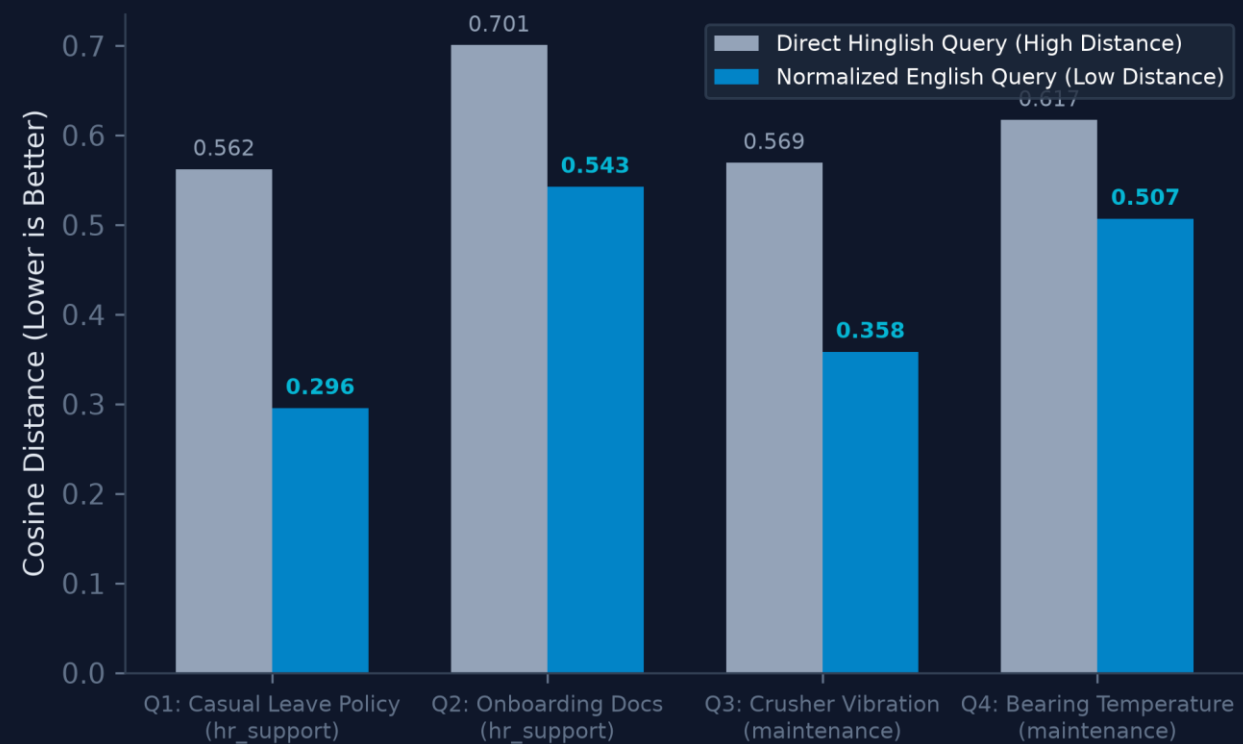
When technicians search in their natural Hinglish syntax, the vector model reports high cosine distance (lower similarity) due to English indexing. By translating Hinglish to English before retrieval, the distance drops significantly, bringing relevant chunks to the top of results.

Key Improvements:

- Q1: Casual Leaves (Distance reduced by 47.4%)
- Q2: Onboarding (Distance reduced by 22.5%)
- Q3: Crusher Vibration (Distance reduced by 37.0%)
- Q4: Bearing Temperature (Distance reduced by 17.8%)

Average improvement across queries is 31.2%, preventing RAG from retrieving wrong files.

Vector Retrieval Distance: Hinglish vs. Translated English
(ChromaDB Multi-lingual Embedding Test)



Query Flow: Specialized RAG Agents

Knowledge & SOP Flow

1. User Query Received
- ↓
2. Role Detection: Checks for specific plant roles (e.g. Battery Operator) in query.
- ↓
3. Collection Targeting: Searches 'role_knowledge' (if role detected), 'knowledge_safety', or 'knowledge_reports' based on keywords.
- ↓
4. Semantic Search: Queries selected vector DB collections.
- ↓
5. Fallback Query Expansion: If no chunks hit, LLM expands query with synonyms (annual report, capex, safety) and retries searching all collections.
- ↓
6. LLM Context Answer: Generates response using context.
- ↓
7. Source Filter: Removes PDF sources if final answer states info is not found in context.

HR Support Flow

1. User Query Received
- ↓
2. HyDE-lite translation: LLM translates query (Hindi/Hinglish) into English.
- ↓
3. Vector Search: Queries 'hr_support' collection using English query.
- ↓
4. Deduplication: Removes duplicate chunks, keeps top scoring results.
- ↓
5. Sufficiency Check (Self-RAG): LLM checks if context answers query. If insufficient, refines query and retries.
- ↓
6. Target Language Matching: Generates final response using original query language (Hindi/Hinglish/English).
- ↓
7. Output: Combines sources and final multilingual answer.

Maintenance Support Flow

1. User Query Received
- ↓
2. Role Scanner: Scans for job roles in query.
- ↓
3. Primary Search: Searches 'maintenance' collection (max distance 0.75).
- ↓
4. Secondary Fallback: If no results, searches 'role_knowledge' collection filtered by detected role metadata.
- ↓
5. Deduplication: Removes duplicate paragraphs.
- ↓
6. Context Synthesis: Merges manual sources.
- ↓
7. Escalation Fallback: If missing, returns response suggesting Maintenance Supervisor (Ext. 301).

Query Flow: Database-Connected SQL Agents

L&D Learning Agent Flow

1. User Query + User ID Received
↓
2. SQLite Connection: Opens data/database.db connection.
↓
3. Profile Extraction: Queries 'users' table for name, department, current/target role, and experience.
↓
4. Skill Gap Retrieval: Queries 'skill_gap' table for missing skills list and readiness score.
↓
5. Course Retrieval: Joins 'recommendations' and 'courses' tables to pull course names, priorities, durations, and skills taught.
↓
6. Task Classification: Scans query keywords to dynamically focus prompt task (duration focus vs. skill gap focus vs. promotion readiness focus).
↓
7. LLM Formulation: Generates structured, personalized roadmap strictly based on SQLite details.

Production KPI Agent Flow

1. User Query Received
↓
2. Orchestrator RBAC authorization check (Employee/New Joiner vs Manager/Executive).
↓ [Access Denied: Returns access blocked response]
3. Access Granted: Orchestrator routes query to Production Agent.
↓
4. Entity Extraction: Extracts query parameters (e.g. Unit number, departments).
↓
5. SQLite SQL query: Queries database for plant efficiency, sensor readings, and downtime statistics (Raw Material Shortage, Power Outages, Equipment Faults).
↓
6. Data Aggregation: Summarizes production KPIs in clean numeric counts.
↓
7. Response Generation: Outputs factual reports and recommendations.

Knowledge Base Structure & Chunking Strategy

Knowledge Base Collections & Source Files

Our RAG database uses ChromaDB with 5 dedicated collections to prevent cross-domain search contamination:

1. `knowledge\_safety` collection
 - File: `safety\_manual.pdf` (Chemical spills, fire, LOTO, PPE codes)
2. `knowledge\_reports` collection
 - Files: `tata\_annual\_report\_2025.pdf`, `sustainability\_2024.pdf` (EBITDA, capex, ESG parameters)
3. `role\_knowledge` collection
 - Files: NSDC role outline PDFs (Heating Regulator, Battery Operator, Control Room, etc.)
4. `maintenance` collection
 - File: `maintenance\_manual.pdf` (Bearing temps, crusher vibrations, PM schedules)
5. `hr\_support` collection
 - Files: `hr\_policy.pdf`, `leave\_policy.pdf`, `onboarding\_doc.pdf`

Chunk Size & Overlap Strategy Rationale

Chunking configurations are tailored specifically to the structural layout of each file type:

- Safety (Size 250 / Overlap 50 words)
Rationale: Keeps sequential step-by-step procedures unified in a single block without truncation.
- Reports (Size 250 / Overlap 50 words)

Rationale: Captures surrounding financial context and comparisons necessary for dense tables or financial metrics.
- Role Knowledge (Size 200 / Overlap 40 words)
Rationale: NSDC PDFs are bullet-heavy. 200 words captures entire individual competency blocks cleanly.
- Maintenance (Size 180 / Overlap 40 words)
Rationale: Keeps troubleshooting guidelines concise to isolate specific errors and limit noise from other machine manual sections.
- HR Support (Size 120 / Overlap 30 words)
Rationale: HR policies contain short standalone rules (e.g. leave days). Tiny chunks pinpoint rules precisely, preventing leakage.

System Demonstration & Screenshots

AI Copilot

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Show my learning progress

18:26

B

Amit, let's review your learning progress:

****Readiness Score:**** Your current score is 33.33, which indicates you are a "Beginner" and need significant development.

****Missing Skills:**** You are missing analytical skills and communication-related skills, which are essential for your target role of Rigger.

****Recommended Courses:****

- **Industrial Communication Skills**** (1 week): Develop your communication skills.
- **Digital Skills for Manufacturing**** (2 weeks): Improve your computer operations and digital reporting skills.
- **Instrumentation & Calibration**** (4 weeks): Enhance your instrumentation and calibration skills.
- **Physical Testing Laboratory Basics**** (2 weeks): Learn physical testing and calibration basics.
- **Control Room Operations**** (10 weeks), ****Rolling Mill Process Control**** (10 weeks), and ****Heating Regulation & Process Monitoring**** (6 weeks) - These courses are also recommended, but let's focus on the most relevant ones first.

****Next Steps:**** Complete the recommended courses and continue developing your analytical and communication skills.

18:26

What skills should I focus on?

What are the safety rules for welding?

What should I learn next?

What happens during probation?

What does a Battery Operator do?

Ask anything...

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Leave policy guidance

18:26

B

Hello! I'm your AI learning assistant. How can I help you today? You can ask me about safety manuals, HR policy, leave approvals, or courses.

18:18

B

As per the leave policy, there are various types of leave available to employees, including Earned Leave, which is encashable upon resignation or retirement. The policy also mentions Special Leave Categories such as Sabbatical Leave, Examination Leave, and Marriage Leave.

To apply for leave, employees must use the HR portal, providing a minimum of 3 days' notice for planned leave. In case of emergency leave, the manager must be informed the same day via call/SMS, and the leave application must be regularized on the portal within 2 days.

Please refer to the leave policy document for detailed information on the leave application process, eligibility, and encashment rules. If you have any further questions or concerns, you can contact HR at hr@tatasteel.com or Extension 201.

18:26

Show my learning progress

B

Amit, let's review your learning progress:

****Readiness Score:**** Your current score is 33.33, which indicates you are a "Beginner" and need

What are the safety rules for welding?

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Ask anything...

System Demonstration & Screenshots

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Cast House Junior Operator Role Description qualifications and training details

20:29

B

The Cast House Junior Operator role involves cleaning/preparation of the cast house, hot metal/slag runners, preparation of mud-gun/tap hole drill machine, making tap hole and metal runners, proper heating etc. before the cast. This also involves tap hole maintenance/preparation, though preparation, metal/slag runner preparation, diamond crossing preparation, mud gun cleaning/filling, drill bar/poking rod changing, spout jam removal, maintain housekeeping, making hook, poking rod, clay spoon, dali etc.

NSQF Level: Minimum Educational Qualifications - Class 8th Pass, Maximum Educational Qualifications - Class 10th Pass. Training includes 2 weeks on-job training, induction training on overview of blast furnace operation, and training on specific job tasks.

Minimum Job Entry Age: 18 years.

Experience: 1-2 years' experience in the similar function or minimum 1 to 3 years' experience as an assistant to a Senior Cast House operator.

Qualifications Pack Code: ISC/Q0406.

Version number: 1.0.

Industry: Iron & Steel.

Drafted on: 12/11/2014.

Last reviewed on: 25/03/2015.

Next review date: 25/03/2016.

What are the safety rules for welding?

What should I learn next?

What happens during probation?

What does a Battery Operator do?

Ask anything...



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What are the safety rules for welding?

18:35

B

For welding, the safety rules are as follows: wear fire retardant clothing, a welding helmet with the appropriate shade, leather gloves, and leather boots/spats.

Additionally, follow the steps outlined in the General Safety Rules section, including wearing Personal Protective Equipment (PPE) at all times and reporting any unsafe conditions immediately.

In the Electrical Safety section, the rule to "never use damaged extension cords or multi-plugs" also applies to welding, as welding equipment often requires electrical power.

What does a Battery Operator do?

18:35

B

A Battery Operator is responsible for general cleaning, housekeeping, battery machine cleaning, preparatory cleaning, technological cleaning of gas carrying equipments, cleaning, sealing and regulation of all sealing surfaces, and operating pollution control equipments. They also prepare ovens before pushing/charging and normalise after pushing/charging. Their tasks include carrying out the sequence of activities for pushing of coke in a coke oven battery, preparing and carrying out the sequence of activities for pushing of coke, and understanding the pushing schedule to plan the sequence of activities.

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18:35

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18:26

What skills should I focus on?

18:26

B

Amit, let's focus on your growth. Your Readiness Score is 33.33, which means you're at a Beginner level and need significant development.

The missing skills that matter are Analytical Skills and Communication-related Skills. Analytical Skills will help you make informed decisions, and Communication-related Skills will improve your collaboration with the team. These skills are crucial for your role as a Rigger: Rigging of Heavy Material, as you'll need to analyze situations and communicate effectively with others.

Considering your priorities, I recommend starting with Industrial Communication Skills and then moving to courses like Instrumentation & Calibration, which can help you develop analytical skills.

18:26

Explain Steel R&D safety rules

18:26

B

No information is available in the retrieved context regarding "Steel R&D safety rules".

System Demonstration & Screenshots

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1. Instrumentation & Calibration (4 weeks) to develop your analytical skills.

2. Instrumentation & Calibration (4 weeks) to develop your analytical skills.

3. Control Room Operations (10 weeks) or Rolling Mill Process Control (10 weeks) can be taken after you've developed your analytical and communication skills.

Next steps: Complete the above courses and reassess your readiness score to plan further development.

18:35

what is the Qualifications Pack for Cast House Junior Operator

20:29

B

The Qualifications Pack for Cast House Junior Operator is ISC/Q0406. It is a set of National Occupational Standards (NOS) that outlines the skills and knowledge required for the job role of a Cast House Junior Operator in the iron and steel industry. The Qualifications Pack includes the following:

- * Job Role: Cast House Junior Operator
- * Sector: Iron & Steel
- * Sub-Sector: Steel
- * Credits (NSQF): TBD
- * Version Number: 1.0
- * Industry: Iron & Steel
- * Drafted on: 12/11/2014
- * Last Reviewed on: 25/03/2015
- * Next Review Date: 25/03/2016
- * NSQC Clearance on: 20/07/2015

The Qualifications Pack is aligned with the National Occupational Standards (NOS) and includes the following units:

- * Understand the assigned job of cast house preparation (ISC/N0422)
- * Understand the preparation job of cast house equipment before the cast (ISC/N0423)

The Qualifications Pack also includes the following Performance Criteria (PC) and marks:

- * ISC/N0422:
 - + PC1: Interpret the log book report for the area to be cleaned (375 marks)
 - + PC2: Plan as appropriate to carry out the housekeeping job as per schedule (45 marks)
 - + PC3: Understand operation of cast house equipment (50 marks)
 - + PC4: Understand blast furnace stove including valve operation (45 marks)
 - + PC5: Understand the functioning of blower, chimney valves, and stove changing (50 marks)
 - + PC6: Understand the specific job and required tackles to be used for preparation of cast house (45 marks)
- * ISC/N0423:
 - + PC1: Ensure proper housekeeping of the working area before opening the cast (375 marks)
 - + PC2: Ensure housekeeping within given time frame to achieve the schedule production target (50 marks)
 - + PC3: Understand the job requirement as per log book report of previous shift (40 marks)
 - + PC4: Ensure cleaning and heating the trough (40 marks)
 - + PC5: Ensure tap hole drill machine and mud gun are kept ready for operation before opening the cast (40 marks)
 - + PC6: Ensure all operating valves of blast furnace stove are in healthy condition (45 marks)

20:29

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